

## SAFETY DATA SHEET

Creation Date 22-September-2009

Revision Date 26-March-2024

Revision Number 3

### 1. Identification

**Product Name** Cadmium oxide, Puratronic®

**Cat No. :** 10664

**CAS-No** 1306-19-0  
**Synonyms** Cadmium oxide; Cadmium oxide fume; Cadmium monoxide

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |             |
|-------------------------------------------------------------|-------------|
| <b>Acute Inhalation Toxicity</b>                            | Category 2  |
| <b>Germ Cell Mutagenicity</b>                               | Category 2  |
| <b>Carcinogenicity</b>                                      | Category 1A |
| <b>Reproductive Toxicity</b>                                | Category 2  |
| <b>Specific target organ toxicity (single exposure)</b>     | Category 3  |
| Target Organs - Respiratory system.                         |             |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 1  |
| Target Organs - Kidney, Blood.                              |             |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Fatal if inhaled  
May cause respiratory irritation  
Suspected of causing genetic defects  
May cause cancer  
Suspected of damaging fertility. Suspected of damaging the unborn child  
Causes damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Immediately call a POISON CENTER/doctor

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component           | CAS-No    | Weight % |
|---------------------|-----------|----------|
| Cadmium oxide (CdO) | 1306-19-0 | >95      |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

**Inhalation**

Remove to fresh air. Immediate medical attention is required. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms/effects** No information available.  
**Notes to Physician** Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Dry chemical, CO<sub>2</sub>, water spray or alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

May be fatal if inhaled. Do not allow run-off from fire-fighting to enter drains or water courses. Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

None known.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
4

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Avoid dust formation.

**Environmental Precautions** Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

**Handling** Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance. Do not get in eyes, on skin, or on clothing. Avoid dust formation.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Acids. Strong oxidizing agents.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component           | Alberta    | British Columbia | Ontario TWA/EV | Quebec     | ACGIH TLV | OSHA PEL | NIOSH                     |
|---------------------|------------|------------------|----------------|------------|-----------|----------|---------------------------|
| Cadmium oxide (CdO) | TWA: 0.002 | TWA: 0.01        | TWA: 0.01      | TWA: 0.025 | TWA: 0.01 |          | IDLH: 9 mg/m <sup>3</sup> |

|  |                   |                                                      |                                                      |                   |                                                      |  |  |
|--|-------------------|------------------------------------------------------|------------------------------------------------------|-------------------|------------------------------------------------------|--|--|
|  | mg/m <sup>3</sup> | mg/m <sup>3</sup><br>TWA: 0.002<br>mg/m <sup>3</sup> | mg/m <sup>3</sup><br>TWA: 0.002<br>mg/m <sup>3</sup> | mg/m <sup>3</sup> | mg/m <sup>3</sup><br>TWA: 0.002<br>mg/m <sup>3</sup> |  |  |
|--|-------------------|------------------------------------------------------|------------------------------------------------------|-------------------|------------------------------------------------------|--|--|

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                              |
|----------------------------------|------------------------------|
| Physical State                   | Powder Solid                 |
| Appearance                       | Red brown                    |
| Odor                             | Odorless                     |
| Odor Threshold                   | No information available     |
| pH                               | No information available     |
| Melting Point/Range              | No data available            |
| Boiling Point/Range              | 1385 °C / 2525 °F @ 760 mmHg |
| Flash Point                      | No information available     |
| Evaporation Rate                 | Not applicable               |
| Flammability (solid,gas)         | No information available     |
| Flammability or explosive limits |                              |
| Upper                            | No data available            |

|                                        |                          |
|----------------------------------------|--------------------------|
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | > 700°C                  |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Cd O                     |
| Molecular Weight                       | 128.41                   |

## 10. Stability and reactivity

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| Reactive Hazard                  | None known, based on information available |
| Stability                        | Stable under normal conditions.            |
| Conditions to Avoid              | Incompatible products. Excess heat.        |
| Incompatible Materials           | Acids, Strong oxidizing agents             |
| Hazardous Decomposition Products | None under normal use conditions           |
| Hazardous Polymerization         | Hazardous polymerization does not occur.   |
| Hazardous Reactions              | None under normal processing.              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Component Information

| Component           | LD50 Oral               | LD50 Dermal | LC50 Inhalation |
|---------------------|-------------------------|-------------|-----------------|
| Cadmium oxide (CdO) | LD50 = 72 mg/kg ( Rat ) | Not listed  | Not listed      |

Toxicologically Synergistic Products No information available

#### Products

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

Irritation No information available

Sensitization No information available

Carcinogenicity Possible cancer hazard. May cause cancer based on animal data. The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component           | CAS-No    | IARC    | NTP   | ACGIH | OSHA | Mexico |
|---------------------|-----------|---------|-------|-------|------|--------|
| Cadmium oxide (CdO) | 1306-19-0 | Group 1 | Known | A2    | X    | A1     |

*IARC (International Agency for Research on Cancer)*

*IARC (International Agency for Research on Cancer)*

*Group 1 - Carcinogenic to Humans*

*Group 2A - Probably Carcinogenic to Humans*

*Group 2B - Possibly Carcinogenic to Humans*

*A1 - Known Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Animal Carcinogen*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*Mexico - Occupational Exposure Limits - Carcinogens*

*Mexico - Occupational Exposure Limits - Carcinogens*

*A1 - Confirmed Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Confirmed Animal Carcinogen*

A4 - Not Classifiable as a Human Carcinogen  
A5 - Not Suspected as a Human Carcinogen

|                                                   |                                                                             |
|---------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Mutagenic Effects</b>                          | Animal experiments showed mutagenic and teratogenic effects                 |
| <b>Reproductive Effects</b>                       | Experiments have shown reproductive toxicity effects on laboratory animals. |
| <b>Developmental Effects</b>                      | No information available.                                                   |
| <b>Teratogenicity</b>                             | No information available.                                                   |
| <b>STOT - single exposure</b>                     | Respiratory system                                                          |
| <b>STOT - repeated exposure</b>                   | Kidney Blood                                                                |
| <b>Aspiration hazard</b>                          | No information available                                                    |
| <b>Symptoms / effects, both acute and delayed</b> | No information available                                                    |
| <b>Endocrine Disruptor Information</b>            | No information available                                                    |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.              |

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Insoluble in water                                                    |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                             |
| <b>Mobility</b>                      | Is not likely mobile in the environment due its low water solubility. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                             |                  |
|-----------------------------|------------------|
| <b>UN-No</b>                | UN2570           |
| <b>Proper Shipping Name</b> | CADMIUM COMPOUND |
| <b>Hazard Class</b>         | 6.1              |
| <b>Packing Group</b>        | I                |

### TDG

|                             |                  |
|-----------------------------|------------------|
| <b>UN-No</b>                | UN2570           |
| <b>Proper Shipping Name</b> | CADMIUM COMPOUND |
| <b>Hazard Class</b>         | 6.1              |
| <b>Packing Group</b>        | I                |

### IATA

|                             |                  |
|-----------------------------|------------------|
| <b>UN-No</b>                | UN2570           |
| <b>Proper Shipping Name</b> | CADMIUM COMPOUND |
| <b>Hazard Class</b>         | 6.1              |
| <b>Packing Group</b>        | I                |

### IMDG/IMO

|                             |                  |
|-----------------------------|------------------|
| <b>UN-No</b>                | UN2570           |
| <b>Proper Shipping Name</b> | CADMIUM COMPOUND |
| <b>Hazard Class</b>         | 6.1              |
| <b>Packing Group</b>        | I                |

## 15. Regulatory information

### International Inventories

| Component           | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|---------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Cadmium oxide (CdO) | 1306-19-0 | X   | -    | X    | ACTIVE                                        | 215-146-2 | -      | -   |

| Component           | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Cadmium oxide (CdO) | 1306-19-0 | X     | KE-04417 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component           | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|---------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Cadmium oxide (CdO) | Part 1, Group B Substance                            |                                                                            |                                           |

### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component           | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                                                         | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)                                             |
|---------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Cadmium oxide (CdO) | -                                                                   | Use restricted. See item 72. (see link for restriction details)<br>Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) Use restricted. See item 23. (see link for restriction details) | SVHC Candidate list - 215-146-2 - Carcinogenic, Article 57a; Specific target organ toxicity after repeated exposure, Article 57(f) - human health |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

#### REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS-No | OECD HPV | Persistent Organic | Ozone Depletion | Restriction of |
|-----------|--------|----------|--------------------|-----------------|----------------|
|-----------|--------|----------|--------------------|-----------------|----------------|

|                     |           |        | Pollutant      | Potential      | Hazardous Substances (RoHS) |
|---------------------|-----------|--------|----------------|----------------|-----------------------------|
| Cadmium oxide (CdO) | 1306-19-0 | Listed | Not applicable | Not applicable | Not applicable              |

| Component           | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Cadmium oxide (CdO) | 1306-19-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y26                      |

## 16. Other information

**Prepared By** Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date** 22-September-2009  
**Revision Date** 26-March-2024  
**Print Date** 26-March-2024  
**Revision Summary** New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**